QORINA - ROUTE OPTIMISATION MESSAGE MAP

How we talk about flight route optimisation

Lead with the decision story: better use of imperfect weather information, compared honestly with the methods people already trust.

Clarify "value chain" first

If someone says value chain, ask which meaning they intend. Here, default to the comparison ladder that shows how value is created: forecast -> trust check -> path choice -> fuel result. This is not the classic business supply chain.

291-555 kg	~97%	~65%	~10 days
saving vs wind-blind on LHR->JFK	exact optimisation capture when fresh	Qorina capture when compared fresh	horizon where learned models are on par

How value is created

The comparison ladder is a decision aid. It does not claim that the model makes the operational decision by itself.

01

Forecast

Start with the wind forecast available before the flight.

02

Trust check

Make uncertainty visible so the recommendation can be judged.

03

Path choice

Compare candidate tracks and select a practical route.

04

Fuel result

The human flight team decides whether to use the recommendation.

Problem

A route can be technically optimal for a forecast that has already gone stale.

The hard question is not only "what is the shortest path?" It is "which path remains sensible when the atmosphere is uncertain?"

That makes forecast age, trust, and comparison choice part of the fuel conversation.

Solution

Qorina learns a route choice from the forecast available at the time.

It adds a trust check, compares against clear yardsticks, and keeps the final choice with the human flight team.

The strongest framing is robust decision support under forecast uncertainty.

Results, with the comparison set visible

LHR->JFK proof of concept - A330-200 - forecast age is the key condition - capture means share of available saving.

Yardstick	Observed result	Why it matters
Wind-blind	291-555 kg saved as the forecast gets fresher	Baseline for explaining why wind-aware routing matters

Exact optimisation on forecast	Wins when the forecast is fresh, about 97% capture	Best fresh-forecast yardstick
MLP / Set Transformer	Roughly on par at about 10-day forecast age	Useful model comparison at the tested horizon
Hindsight optimum	Upper bound using information known after the fact	Ceiling, not a live operating target

Source: Qorina flight-routing PoC comparison - indicative route-level evidence, not a network-wide operating result.

When the story is strong

The forecast is imperfect, ageing, or materially uncertain.

Qorina gives a stable, inspectable recommendation and makes the tradeoff easy to compare.

The useful message is: value comes from making a good decision with the information actually available.

When the story is weaker

The forecast is fresh and exact optimisation has the strongest information advantage.

In that setting, position Qorina as a comparison point and decision-support layer, not the automatic winner.

The honest message is: performance depends on forecast age.

One-sentence version

Qorina turns the forecast available before departure into a trust-checked path choice, then shows the fuel result against the right comparison ladder.

Do not say

- Proven airline savings
- Beats exact optimisation always
- Quantum advantage
- Ready to replace certified flight plans